Loss-Based Client Quality Estimation for Robust Federated Learning under Label Noise

***Abstract*—Federated Learning (FL) enables collaborative model training without sharing raw data, making it attractive for privacy-sensitive medical image analysis. However, non-IID data and annotation corruption can produce unreliable client updates and degrade global optimization. Existing contribution-aware approaches often rely on computationally expensive valuation or similarity-based signals that may become unreliable under partial participation. This work proposes a lightweight loss-based client quality estimator integrated with FedDyn. The broadcast global model's prediction loss on each participating client's data is converted into a quality score, blended with the client-size prior, and bounded to derive aggregation weights. The same final weights are used consistently for model aggregation and the FedDyn dynamic correction. Experiments on OCTMNIST, ISIC, and APTOS under a frozen non-IID protocol show competitive clean-data performance. Mechanism analysis on OCTMNIST further shows that corrupted clients receive approximately $0.69\times$ the aggregation weight of clean clients. Yet, down-weighting does not provide a general robustness advantage over FedDyn: it approaches parity where corruption degrades FedDyn most, but is less effective where the baseline is already noise-tolerant. The study also documents a failure mode of cosine-based contribution scoring under partial participation and highlights the distinction between reliable client detection and global optimization improvement.**

***Index Terms*—Federated learning, client quality estimation, label noise, FedDyn, non-IID data, medical image classification**

I. Introduction

Federated Learning (FL) enables multiple institutions to collaboratively train machine learning models while retaining sensitive data locally [1], [2]. This paradigm is particularly relevant to healthcare, where patient data are distributed across hospitals and centralized collection may be restricted by privacy, governance, and regulatory requirements. By exchanging model information rather than raw records, FL provides a practical foundation for collaborative medical image analysis.

Real-world federated learning remains challenging because participating institutions rarely contain identically distributed data. Differences in patient populations, imaging devices, acquisition protocols, disease prevalence, and annotation practices produce statistically heterogeneous or non-IID client datasets [3]. Such heterogeneity can cause local optimization trajectories to diverge from the global objective, leading to client drift and unstable convergence.

Several federated optimization methods address this problem. FedProx introduces proximal regularization to constrain local optimization [4], SCAFFOLD uses control variates to correct client drift [5], and FedDyn employs dynamic regularization to improve consistency between local and global objectives [6]. These methods improve optimization under heterogeneity but do not explicitly estimate whether individual client updates are reliable.

Reliability is particularly important in medical learning because local annotations may contain errors arising from inter-observer variability, ambiguous cases, or inconsistent labeling protocols. A client trained on corrupted labels may return an update that is locally optimized but harmful to the global model. Conventional size-based aggregation may assign such a client substantial influence simply because it owns more data.

Contribution-aware FL attempts to address unequal client usefulness through client-specific valuation. Shapley-value-based methods provide a principled framework for estimating marginal contribution but may require expensive coalition evaluation [7], [8]. Lower-cost alternatives use gradient alignment or update similarity [9], [10]. However, similarity-based signals depend on a reference update direction that may itself be contaminated when participation is partial and multiple corrupted clients are active.

Motivated by these limitations, we investigate whether prediction loss can serve as a lightweight client-reliability signal. We propose a loss-based quality estimator integrated with FedDyn. Client quality is computed from the broadcast global model's prediction loss on each participating client's local data. The quality score is blended with the sample-size prior and bounded before aggregation. The same final weights are used consistently in both model aggregation and the FedDyn dynamic correction.

The main contributions are:

- A lightweight loss-based client-quality estimator that avoids combinatorial Shapley valuation and pairwise update comparison.
- A weight-consistent integration with FedDyn in which the same quality-aware coefficients are used for model aggregation and dynamic correction.
- A mechanism-level validation on OCTMNIST showing that corrupted clients receive approximately $0.69\times$ the aggregation weight of clean clients under the evaluated noisy-client setting.
- A documented negative result showing that cosine-based contribution scoring can become anti-discriminative under partial participation.
- A multi-dataset characterization showing that the relevance of down-weighting depends on how strongly corruption degrades the underlying FedDyn baseline.

II. Related Work

A. Federated Optimization under Heterogeneity

FedAvg established the standard FL paradigm in which clients perform local optimization and the server aggregates their updates using data-size-based weights [1]. Although communication-efficient, FedAvg may experience client drift under heterogeneous local data.

FedProx introduces a proximal term that constrains local models relative to the global model [4]. SCAFFOLD uses control variates to correct local drift [5], while FedDyn introduces dynamic regularization to align local and global optimization [6]. The present work therefore uses FedDyn as the optimization backbone and primary optimization baseline.

B. Contribution and Quality-Aware FL

Client contribution estimation has been investigated for valuation, selection, fairness, and aggregation. Shapley-based approaches estimate marginal utility across client coalitions [7], [8], but exact or repeated coalition evaluation can become computationally expensive.

Alternative methods use lower-cost proxy signals. FedCE considers contribution estimation in federated medical imaging using update-related information [9], while ShapFed uses a cosine-similarity based contribution signal for federated weighting [10]. Such signals reduce valuation cost but depend on the reliability of the reference update direction.

Under partial participation and label corruption, several unreliable clients may simultaneously influence the active-set consensus. This can make update-alignment signals misleading. We therefore focus on client *reliability* rather than combinatorial contribution and use prediction loss as an independent, lightweight quality proxy.

III. Problem Formulation

Consider a cross-silo FL system with a central server and N clients, $\mathcal{C} = \{1, \dots, N\}$. Client i owns private dataset D_i with n_i samples. The global objective is

$$\min_W F(W) = \sum_{i=1}^{N} p_i F_i(W), \qquad p_i = \frac{n_i}{\sum_{j=1}^{N} n_j}, \tag{1}$$

where $F_i(W)$ is the empirical objective of client i, n_i denotes its number of samples, and p_i is the corresponding normalized size prior.

At communication round t, the server selects $S_t \subseteq \mathcal{C}$ and broadcasts global model W^t. Each selected client performs local FedDyn training and returns updated parameters W_i^t.

Under heterogeneous and corrupted local supervision, client updates may differ substantially in reliability. Our objective is to estimate a client-specific quality signal and reduce the influence of potentially unreliable clients while preserving useful information associated with client data volume.

The global update can be written as

$$W^{t+1} = \sum_{i \in S_t} \alpha_i^t W_i^t, \qquad \sum_{i \in S_t} \alpha_i^t = 1, \tag{2}$$

where α_i^t denotes the final aggregation weight assigned to client i at round t.

IV. Proposed Method

The proposed framework augments FedDyn with a loss-based client-quality detector, bounded quality-aware weighting, and weight-consistent FedDyn aggregation. Fig. 1 summarizes one communication round.

A. Local FedDyn Training and Loss Evaluation

At round t, five participating clients are selected randomly from the ten available clients. The server broadcasts the current global model W^t to all selected clients.

Each client performs local FedDyn optimization starting from W^t to obtain its updated local model W_i^t. The FedDyn local objective is retained unchanged [6]; hence, the proposed contribution concerns client quality estimation and aggregation rather than modification of the local optimizer.

For quality estimation, the broadcast global model W^t is evaluated on client i's local dataset D_i to obtain the average cross-entropy loss

$$L_i^t = \frac{1}{|D_i|} \sum_{(x,y) \in D_i} \ell\left(W^t; x, y\right), \tag{3}$$

where $\ell(\cdot)$ denotes the cross-entropy loss.

Thus, L_i^t reflects how well the current shared global model agrees with the supervision available at client i. Only the local model parameters and the scalar loss statistic are communicated; raw client data remain local.

B. Loss-Based Client-Quality Estimation

Prediction loss is used as a soft reliability proxy. Corrupted supervision can increase disagreement between local labels and predictions of the evolving global model, leading to larger loss. Since loss may also reflect intrinsic task difficulty or distribution shift, it is not treated as a hard corruption detector.

For participating client i, the quality score is

$$Q_i^t = \exp\left[-\beta\left(L_i^t - L_{\min}^t\right)\right], \tag{4}$$

where

$$L_{\min}^t = \min_{j \in S_t} L_j^t, \tag{5}$$

and $\beta > 0$ controls sensitivity to loss differences.

The shifted formulation improves numerical stability while preserving the relative ordering of client-quality scores.

The quality score is normalized as

$$\hat{Q}_i^t = \frac{Q_i^t}{\sum_{j \in S_t} Q_j^t}. \tag{6}$$

Thus, lower-loss clients receive larger normalized quality scores, while higher-loss clients receive reduced influence.

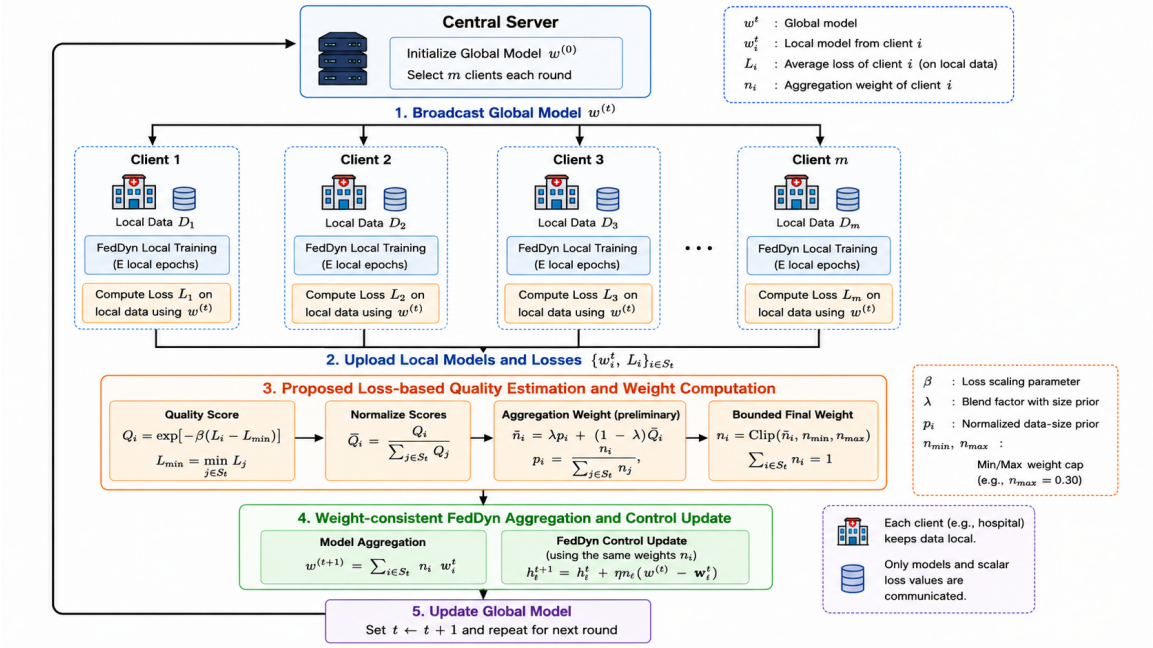


Fig. 1. Proposed loss-based quality-aware FedDyn framework. Selected clients perform local FedDyn training and return their updated models and prediction losses. Loss-based quality scores are combined with the client-size prior to derive bounded aggregation weights, which are used consistently for global model aggregation and the FedDyn dynamic correction.

C. Blended and Bounded Quality Weights

Pure quality weighting can over-concentrate aggregation on a small number of clients. To retain the contribution of data quantity, the normalized quality score is combined with the client-size prior.

For participating clients, the round-specific size prior is

$$p_i^t = \frac{n_i}{\sum_{j \in S_t} n_j}. \tag{7}$$

The preliminary aggregation coefficient is

$$\tilde{\alpha}_i^t = \lambda p_i^t + (1 - \lambda)\hat{Q}_i^t, \tag{8}$$

where $\lambda \in [0, 1]$ controls the balance between data quantity and estimated reliability.

To prevent a single client from dominating aggregation, the preliminary weights are bounded using a maximum weight $w_{\max}$ and subsequently renormalized:

$$\bar{\alpha}_i^t = \min\left(\tilde{\alpha}_i^t, w_{\max}\right), \tag{9}$$

$$\alpha_i^t = \frac{\bar{\alpha}_i^t}{\sum_{j \in S_t} \bar{\alpha}_j^t}. \tag{10}$$

The final configuration uses

$$\beta = 1.5, \qquad \lambda = 0.5, \qquad w_{\max} = 0.30. \tag{11}$$

These moderate settings preserve participation diversity while reducing the influence of high-loss clients.

D. Weight-Consistent FedDyn Integration

The global model is aggregated using the final coefficients:

$$W^{t+1} = \sum_{i \in S_t} \alpha_i^t W_i^t. \tag{12}$$

An important implementation detail is that the same coefficients α_i^t are also used in the corresponding FedDyn dynamic correction. This avoids a mismatch in which reliability-aware coefficients are used for model aggregation while a different weighting scheme is retained in the dynamic regularization component. Therefore, client reliability influences both components consistently. Under the standard weighting configuration, the implementation reduces to the corresponding standard FedDyn behavior.

E. Design Rationale

Random client selection is used instead of loss-driven Power-of-Choice selection because high-loss clients under label corruption may be precisely those containing unreliable annotations. Preferentially selecting such clients could increase rather than reduce their influence. Loss is preferred over cosine-based update similarity because, under partial participation, multiple corrupted clients can shift the reference update direction toward unreliable updates. A corrupted update may then appear strongly aligned with a consensus that it helped contaminate. Finally, moderate rather than aggressive quality weighting is used. Experiments with stronger sensitivity, $\beta = 4$ and weight cap 0.5, caused aggregation to concentrate on one or two clients and produced unstable late-round behavior.

TABLE I
DATASETS AND FROZEN EXPERIMENTAL CONFIGURATION.

Datasets		FL Configuration	
OCTMNIST	4 classes	Clients	10
ISIC	7 classes	Active/round	5
APTOS-2019	5 classes	Local epochs	2
Model	ResNet-18	Rounds	40
Input	64×64	Dirichlet α	0.3
Optimizer	SGD	Initial LR	10^{-3}

F. Training Procedure

Algorithm 1 summarizes the proposed training process. The additional quality-weight computation requires only $O(|S_t|)$ scalar operations beyond loss evaluation. It avoids combinatorial coalition enumeration and pairwise update comparison.

Algorithm 1 Loss-Based Quality-Aware FedDyn

Require: Clients $\mathcal{C}$, rounds T, global model W^0
for $t = 0, \ldots, T-1$ **do**
  Randomly select participating clients S_t
  Broadcast global model W^t
  for all $i \in S_t$ **do**
    Evaluate W^t on D_i to obtain loss L_i^t
    Perform local FedDyn training from W^t
    Obtain updated local model W_i^t
  end for
  Compute $L_{\min}^t = \min_{i \in S_t} L_i^t$
  for all $i \in S_t$ **do**
    $Q_i^t \leftarrow \exp[-\beta(L_i^t - L_{\min}^t)]$
  end for
  Normalize quality scores to obtain $\hat{Q}_i^t$
  Compute size priors p_i^t
  Compute $\tilde{\alpha}_i^t = \lambda p_i^t + (1-\lambda)\hat{Q}_i^t$
  Apply weight cap $w_{\max}$
  Renormalize to obtain final weights α_i^t
  Aggregate $W^{t+1} = \sum_{i \in S_t} \alpha_i^t W_i^t$
  Update FedDyn correction using the same α_i^t
end for
return W^T

V. EXPERIMENTAL SETUP

A. Datasets and Frozen Protocol

Three medical image classification tasks are evaluated.

OCTMNIST is a four-class retinal optical coherence tomography dataset from the MedMNIST benchmark [11]. ISIC is used for seven-class dermoscopic skin-lesion classification and is derived from the public ISIC archive [12]. APTOS-2019 is a five-grade ordinal diabetic retinopathy classification task based on retinal fundus images.

All datasets are partitioned non-IID using a Dirichlet distribution with $\alpha = 0.3$. The same frozen training and evaluation protocol is used for all compared methods.

Table I summarizes the main configuration.

An ImageNet-pretrained ResNet-18 is trained using SGD with initial learning rate 10^{-3} and cosine learning-rate decay. Each selected client performs two local epochs, and training continues for 40 communication rounds. The training set is capped at 20,000 samples for consistent computational cost. Metrics are read at the final converged round rather than selecting the best intermediate round.

B. Noisy-Client Model

Two conditions are evaluated here. clean training and a noisy-client setting. In the noisy setting, 40% of clients are designated as corrupted. For each corrupted client, 60% of its local training labels are changed to a random incorrect class. Test labels remain clean, and the same corrupted client set is used across compared methods for a given seed. Thus, the noisy-client setting denotes 40% corrupted clients, each with 60% of its local training labels flipped; it does not denote corruption of 40% of all training labels.

C. Baselines and Evaluation Metrics

FedAvg is used as the conventional size-weighted baseline, while FedDyn is used as the primary optimization baseline and the optimization backbone of the proposed method. The same data partitions, client-participation protocol, optimizer settings, and corrupted-client assignments are used across comparisons. Global test accuracy is used as the primary predictive metric. Jain's fairness index measures the evenness of client-level predictive performance. Aggregation weights are additionally analyzed to determine whether the proposed quality estimator distinguishes clean and corrupted clients.

VI. RESULTS AND DISCUSSION

A. Performance under Clean Non-IID Data

The comparison on FedAvg, FedDyn, and the proposed method under clean non-IID conditions are depicted in Fig. 2

On OCTMNIST, FedAvg obtains 0.636 ± 0.057, FedDyn reaches 0.688 ± 0.035, and the proposed method achieves 0.708 ± 0.029. On APTOS, the corresponding accuracies are 0.678 ± 0.011, 0.715 ± 0.023, and 0.728 ± 0.014. On ISIC, FedDyn obtains the highest mean accuracy of 0.734 ± 0.013, while the proposed method reaches 0.719 ± 0.019. The OCTMNIST and APTOS differences relative to FedDyn are small compared with seed variability and should therefore be interpreted as competitive parity rather than definitive superiority.

Introducing client-quality weighting preserves competitive clean-data performance while adding reliability awareness to the FedDyn backbone.

B. Robustness under Noisy Clients

The performance under the noisy-client setting .is depicted in Table II. On OCTMNIST, FedDyn obtains 0.523 ± 0.075, while the proposed method reaches 0.518 ± 0.042. The difference is small relative to the observed cross-seed variability. On ISIC and APTOS, FedDyn remains stronger in mean accuracy. Therefore, down-weighting unreliable clients does not provide a general robustness advantage over an already strong dynamically regularized optimizer.

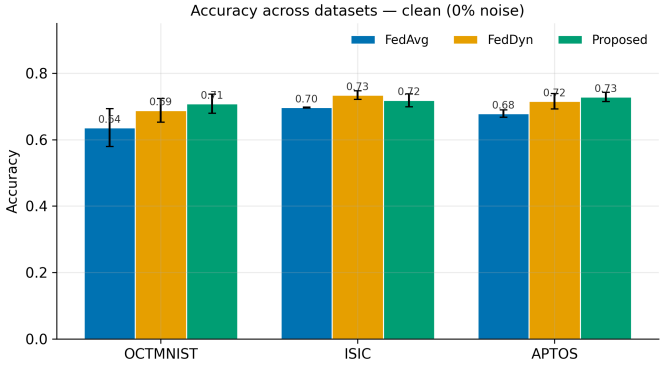


Fig. 2. Global test accuracy of FedAvg, FedDyn, and the proposed method under clean non-IID settings. Error bars denote variability across independent runs.

TABLE II
PERFORMANCE WITH 40% CORRUPTED CLIENTS.

Data	Method	Accuracy	Jain
OCT	FedAvg	$.462 \pm .054$	.885
	FedDyn	$\mathbf{.523 \pm .075}$	**.926**
	Proposed	$.518 \pm .042$	.904
ISIC	FedAvg	$.669 \pm .007$	.703
	FedDyn	$\mathbf{.672 \pm .026}$	**.825**
	Proposed	$.645 \pm .046$	.809
APTOS	FedAvg	$.577 \pm .083$	.716
	FedDyn	$\mathbf{.649 \pm .084}$	**.816**
	Proposed	$.588 \pm .133$	**.816**

Reducing the influence of a corrupted client can suppress harmful supervision, but the same client may still contain useful non-IID information. Reliability-aware weighting therefore creates a trade-off between robustness and preservation of complementary data.

C. Mechanism Validation

Predictive accuracy alone cannot establish whether the proposed quality estimator behaves as intended. We therefore directly analyze the aggregation weights assigned to clean and corrupted clients.

On OCTMNIST, Fig. 3 shows that clean clients receive a mean aggregation weight of approximately 0.228, whereas corrupted clients receive approximately 0.156. The corrupted-to-clean weight ratio is

$$\frac{0.156}{0.228} \approx 0.69. \tag{13}$$

.The separation persists throughout training, confirming that the proposed mechanism systematically reduces the influence of corrupted clients.

However, mechanism success and optimization success are distinct. Correctly identifying unreliable clients does not guarantee higher global accuracy, because optimization also depends on data diversity, class distribution, sample size, participation, and baseline optimizer robustness.

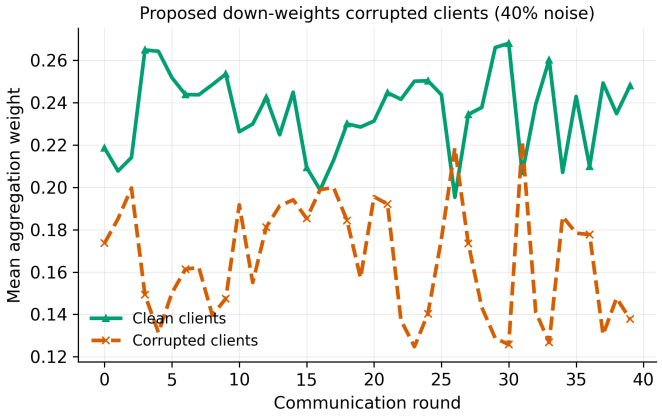


Fig. 3. Mean aggregation weights assigned to clean and corrupted clients on OCTMNIST with 40% corrupted clients. Corrupted clients receive approximately $0.69\times$ the weight of clean clients.

TABLE III
FEDDYN DEGRADATION AND PROPOSED-METHOD GAP UNDER NOISY CLIENTS. NEGATIVE GAPS INDICATE HIGHER ACCURACY FOR FEDDYN.

Dataset	FedDyn Opening	Proposed − FedDyn
OCTMNIST	0.158	−0.006
APTOS	0.066	−0.061
ISIC	0.062	−0.027

D. Negative Result: Cosine-Based Contribution Scoring

We additionally evaluated cosine-of-update contribution scoring following the similarity-based contribution principle used by methods such as ShapFed [10].Under partial participation, this signal became anti-discriminative on OCTMNIST. Across three evaluated seeds, corrupted clients received a higher contribution scores and greater aggregation influence, with corrupted-to-clean weight ratios of approximately 1.1–1.3. A likely explanation is contamination of the reference direction. With five of ten clients participating and 40% of clients corrupted, an active set can contain multiple corrupted participants. Their updates shift the consensus direction, causing corrupted updates to appear aligned with a reference they helped create. This negative result motivated the use of prediction loss, which evaluates each client's consistency without requiring comparison against a potentially contaminated consensus update.

E. When Does Down-Weighting Become Relevant?

To interpret the dataset-dependent behavior, we define the *baseline opening* as the reduction in FedDyn accuracy from clean to the noisy-client setting:

$$\Delta_{\text{open}} = A^{\text{clean}}_{\text{FedDyn}} - A^{\text{noise}}_{\text{FedDyn}}. \tag{14}$$

Table III compares this opening with the proposed method's accuracy difference relative to FedDyn under noisy clients.

OCTMNIST exhibits the largest degradation of FedDyn and is also the dataset where the proposed method comes

closest to FedDyn. ISIC and APTOS exhibit smaller baseline degradation and larger negative gaps.

The results suggest that reliability-aware down-weighting becomes more relevant as corruption increasingly degrades the underlying optimizer, although this does not imply that it will necessarily outperform the baseline. When FedDyn already absorbs moderate corruption effectively, there is limited performance degradation to recover, while reducing client influence may discard useful information.

F. Weighting Stability

The strength of quality weighting affects optimization stability. Aggressive weighting with $\beta = 4$ and a weight cap of 0.5 concentrated aggregation on one or two clients and caused late-round instability, with some runs falling to approximately 0.27–0.39 final accuracy despite stronger intermediate performance.The final moderate configuration,

$$\beta = 1.5, \qquad \lambda = 0.5, \qquad w_{\max} = 0.30,$$

substantially reduced this collapse.

This supports soft rather than aggressive client suppression. A high-loss client may contain corrupted labels, but it may also represent difficult, rare, or distributionally distinct data.

G. Discussion and Limitations

Three main observations emerge from the study. First, FedDyn is a strong optimization baseline under both clean and corrupted non-IID settings. Reliability-aware aggregation should therefore be evaluated against strong federated optimizers rather than only FedAvg. Second, the proposed loss-based estimator successfully reduces corrupted-client influence on OCTMNIST. Prediction loss therefore provides a simple and interpretable reliability signal in the evaluated setting.

Third, reliable detection does not imply a universal accuracy gain. The benefit depends on how strongly unreliable clients degrade the underlying optimizer and on how much useful information is lost when their influence is reduced. The study considers only clean and one corrupted-client regime, focuses on label corruption rather than Byzantine model manipulation or feature shift, and uses a single participation configuration with ten clients and five active per round.

Prediction loss may also reflect class imbalance, intrinsic task difficulty, or distribution shift in addition to annotation reliability. Future work should therefore investigate multi-signal quality estimators combining loss with temporal consistency, uncertainty, validation utility, or historical update stability.

VII. Conclusion

This work presented a lightweight loss-based client-quality estimator for robust federated learning under non-IID data and label corruption. The method integrates with FedDyn by converting the broadcast global model's prediction loss on client data into bounded quality-aware aggregation weights. These weights are blended with the sample-size prior and used consistently for model aggregation and the FedDyn dynamic correction.

Experiments on OCTMNIST, ISIC, and APTOS show competitive clean-data performance. Mechanism analysis on OCTMNIST verifies that corrupted clients receive approximately $0.69\times$ the aggregation influence of clean clients. However, down-weighting does not provide a general robustness advantage over FedDyn under the evaluated noisy-client regime.

The study additionally identifies a failure mode of cosine-based contribution scoring under partial participation, where a contaminated consensus can make corrupted clients appear highly aligned. The results emphasize that reliable client-quality estimation and improved global optimization are related but distinct objectives. Future work will investigate multi-signal reliability estimation, adaptive weighting, broader corruption regimes, and larger cross-silo federated learning deployments.

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